

5.5

$$a. \text{var}(b_2|x) = \frac{\sigma^2}{\sum_{t=1}^T (x_t - \bar{x})^2}$$

$$\sum_{t=1}^T (t - \bar{t})^2 = \sum_{t=1}^T t^2 - \frac{(\sum_{t=1}^T t)^2}{T}$$

$$\text{其中, } \sum_{t=1}^T t^2 = \frac{T(T+1)(2T+1)}{6}, \quad \sum_{t=1}^T t = \frac{T(T+1)}{2}$$

$$\begin{aligned} \Rightarrow \sum_{t=1}^T (t - \bar{t})^2 &= \frac{T(T+1)(2T+1)}{6} - \frac{T^2(T+1)^2}{4T} = \frac{T(T+1)}{2} \left[\frac{2T+1}{3} - \frac{(T+1)}{2} \right] \\ &= \frac{T(T^2-1)}{12} \end{aligned}$$

$$\Rightarrow \text{Var}(b_2|x) = \frac{\sigma^2}{\frac{T(T^2-1)}{12}} = \frac{12\sigma^2}{T(T^2-1)}$$

$\Rightarrow \lim_{T \rightarrow \infty} \text{Var}(b_2|x) = 0$, the least square estimator b_2 is consistent

$$b. \sum_{t=1}^T (x_t - \bar{x})^2 = \sum_{t=1}^T x_t^2 - \frac{(\sum_{t=1}^T x_t)^2}{T} = \frac{1 - (1/4)^T}{3} - \frac{(1 - (1/2)^T)^2}{T}$$

$$\sum_{t=1}^T x_t = \sum_{t=1}^T \left(\frac{1}{2}\right)^t = \frac{\frac{1}{2}(1 - (1/2)^T)}{1 - 1/2} = 1 - (1/2)^T$$

$$\sum_{t=1}^T x_t^2 = \sum_{t=1}^T \left(\frac{1}{4}\right)^t = \frac{\frac{1}{4}(1 - (1/4)^T)}{1 - 1/4} = \frac{1 - (1/4)^T}{3}$$

$$\text{Var}(b_2|x) = \sigma^2 / \left(\frac{1 - (1/4)^T}{3} - \frac{(1 - (1/2)^T)^2}{T} \right)$$

$$\lim_{T \rightarrow \infty} \text{Var}(b_2|x) = 3\sigma^2 \neq 0$$

the least square estimator b_2 is not consistent

C.

a. : Good Scenario, Because x values spread further apart over time, the model has a very clear map to follow

b. : Bad Scenario, Because x values quickly get stuck near zero. Adding more data doesn't help.

5.6

a. $H_0: \beta_2 = 0$, $H_1: \beta_2 \neq 0$

$$\alpha = 0.05$$

$$t = \frac{b_2 - 0}{\text{se}(b_2)} \overset{H_0}{\sim} t(60)$$

$$t^* = \frac{3}{2} = 1.5$$

$$RR = \{t \mid |t| \geq t_{0.015}(60) = 2\}$$

$t^* \notin RR$, do not reject H_0 , we don't have enough evidences to say $\beta_2 \neq 0$

b. $H_0: \beta_1 + 2\beta_2 = 5$. $H_1: \beta_1 + 2\beta_2 \neq 5$

$$\alpha = 0.05$$

$$t = \frac{b_1 + 2b_2 - 5}{\text{se}(b_1 + 2b_2)} \overset{H_0}{\sim} t(60) \quad \text{se}(b_1 + 2b_2) = \sqrt{\text{Var}(b_1) + 4\text{Var}(b_2) + 4\text{Cov}(b_1, b_2)}$$

$$t^* = \frac{3}{\sqrt{11}} \approx 0.9045 \quad = \sqrt{3 + 16 - 8} = \sqrt{11}$$

$$RR = \{t \mid |t| \geq t_{0.025}(60) = 2\}$$

$t^* \notin RR$, do not reject H_0 , we don't have enough evidences to say $\beta_1 + 2\beta_2 \neq 5$

c. $H_0: \beta_1 - \beta_2 + \beta_3 = 4$, $H_1: \beta_1 - \beta_2 + \beta_3 \neq 4$

$$\alpha = 0.05$$

$$t = \frac{b_1 - b_2 + b_3 - 4}{\text{se}(b_1 - b_2 + b_3)} \overset{H_0}{\sim} t(60) \quad \text{se}(b_1 - b_2 + b_3) = \sqrt{\text{Var}(b_1) + \text{Var}(b_2) + \text{Var}(b_3) + 2(\text{cov}(b_1, b_2) + \text{cov}(b_3, b_2) + \text{cov}(b_1, b_3))} = 4$$

$$t^* = \frac{-2-4}{4} = -1.5$$

$$RR = \{t \mid |t| \geq t_{0.025}(60) = 2\}$$

$t^* \notin RR$, do not reject H_0 , we don't have enough evidences to say $\beta_1 - \beta_2 + \beta_3 \neq 4$

5.2

a. $\bar{y}_1 = \beta_1 + \beta_2 \bar{x}_1 + \bar{e}_1$

$\bar{y}_2 = \beta_1 + \beta_2 \bar{x}_2 + \bar{e}_2$

$$\hat{\beta}_{2, \text{mean}} = \frac{(\beta_1 + \beta_2 \bar{x}_2 + \bar{e}_2) - (\beta_1 + \beta_2 \bar{x}_1 + \bar{e}_1)}{\bar{x}_2 - \bar{x}_1} = \beta_2 + \frac{\bar{e}_2 - \bar{e}_1}{\bar{x}_2 - \bar{x}_1}$$

$\therefore E(\bar{e}_i | x_i) = 0$, then $E(\bar{e}_2 - \bar{e}_1 | x) = 0$

$\therefore E(\hat{\beta}_{2, \text{mean}}) = \beta_2 \rightarrow$ is an unbiased estimator

b. $\text{Var}(\hat{\beta}_{2, \text{mean}} | x) = \text{Var}(\beta_2 + \frac{\bar{e}_2 - \bar{e}_1}{\bar{x}_2 - \bar{x}_1} | x) = \frac{1}{(\bar{x}_2 - \bar{x}_1)^2} \text{Var}(\bar{e}_2 - \bar{e}_1 | x)$

$$= \frac{1}{(\bar{x}_2 - \bar{x}_1)^2} \frac{4\sigma^2}{N} = \frac{4\sigma^2}{N(\bar{x}_2 - \bar{x}_1)}$$

$\therefore \text{Var}(\bar{e}_i | x) = \sigma^2$

$\Rightarrow \text{var}(\bar{e}_1 | x) = \text{var}(\frac{2}{N} \sum e_i | x) = (\frac{2}{N})^2 \cdot (\frac{N}{2}) \sigma^2 = \frac{2\sigma^2}{N}$

$\therefore \text{Var}(\bar{e}_2 - \bar{e}_1 | x) = \text{var}(\bar{e}_1 | x) + \text{var}(\bar{e}_2 | x) = \frac{4\sigma^2}{N}$

c. $\text{Var}(\hat{\beta}_{2, \text{mean}}) = E(\text{Var}(\hat{\beta}_{2, \text{mean}})) + \text{Var}(E(\hat{\beta}_{2, \text{mean}}))$

$$= E\left(\frac{4\sigma^2}{N(\bar{x}_2 - \bar{x}_1)^2}\right) + 0$$

$$= \frac{4\sigma^2}{N} E\left[\frac{1}{(\bar{x}_2 - \bar{x}_1)^2}\right]$$

d. $\hat{\beta}_{2, \text{mean}} \xrightarrow{P} \beta_2$, when $N \rightarrow \infty$

if $\text{Var}(\hat{\beta}_{2, \text{mean}}) \rightarrow 0$, then $\frac{1}{N} E\left[\frac{1}{(\bar{x}_2 - \bar{x}_1)^2}\right] \rightarrow 0$ - ①

if $\bar{x}_2 - \bar{x}_1 \xrightarrow{P} c$, where $c \neq 0$, then $E\left[\frac{1}{(\bar{x}_2 - \bar{x}_1)^2}\right] \rightarrow \frac{1}{c^2}$ - ②

by ①, ②, under ② $E\left[\frac{1}{(\bar{x}_2 - \bar{x}_1)^2}\right] \rightarrow 0$, as $N \rightarrow \infty$

$\text{Var}(\hat{\beta}_{2, \text{mean}}) \rightarrow 0$, then $\hat{\beta}_{2, \text{mean}} \xrightarrow{P} \beta_2$

e.

N <dbl>	var_b2_OLS <dbl>	var_beta2_mean <dbl>	estimate_inv_sx2 <dbl>
100	1.30041194	1.70041906	0.1300412
500	0.24566178	0.33214099	0.1228309
1000	0.11947361	0.16006828	0.1194736
5000	0.02420016	0.03238621	0.1210008

f. OLS is more more efficient than the mean estimator because it has a smaller var, meaning its results are more precise and stable. While both methods become more accurate as you collect more data, OLS stays more reliable throughout. Since the error for both eventually skinks to 0 with enough data, there both consistent estimators.

$$g. E[(S_x^2)^{-1}] \xrightarrow{P} 0.12$$

$$\because X \sim U(0,10)$$

$$\therefore \text{Var}(X) = \frac{(b-a)^2}{12} = \frac{25}{3}$$

$$\therefore S_x^2 = \frac{1}{N} \sum_{i=1}^N (X_i - \bar{X})^2$$

$$\therefore S_x^2 \xrightarrow{P} \text{Var}(X)$$

$$\therefore (S_x^2)^{-1} \xrightarrow{P} \left(\frac{25}{3}\right)^{-1} = 0.12$$

$$\therefore E[(S_x^2)^{-1}] \rightarrow 0.12$$

$$E[4/(\bar{X}_2 - \bar{X}_1)^2] \xrightarrow{P} 0.16$$

$$\because \bar{X}_2 \xrightarrow{P} 7.5, \bar{X}_1 \xrightarrow{P} 2.5$$

$$\therefore \bar{X}_2 - \bar{X}_1 \xrightarrow{P} 5$$

$$\therefore \frac{4}{(\bar{X}_2 - \bar{X}_1)^2} \xrightarrow{P} \frac{4}{25}$$

$$\therefore E[4/(\bar{X}_2 - \bar{X}_1)^2] \xrightarrow{P} 0.16$$

h. No, it will generally not be consistent.

Because if $\bar{X}_2 - \bar{X}_1 \rightarrow 0$, $\hat{\beta}_{2, \text{mean}}$ will be unstable, which means that $\hat{\beta}_{2, \text{mean}} \not\rightarrow \beta_2$